CHAPTER 4

CHEMICAL BONDING AND MOLECULAR STRUCTURE

CHAPTER ESSENCE AT A GLANCE:
Explains chemical bonding models, VSEPR molecular geometry, atomic orbital hybridization, Molecular Orbital Theory, and hydrogen bonding.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **VSEPR Molecular Geometry:** Electron pairs arrange symmetrically around the central atom to minimize repulsive forces: lp-lp > lp-bp > bp-bp.

• **Orbital Hybridization:** Intermixing of atomic orbitals produces equivalent directional hybrid orbitals defining molecular shape and bond angles.

• **Molecular Orbital Theory (MOT):** LCAO combination creates bonding and antibonding molecular orbitals; bond order dictates stability and bond length.

• **Hydrogen Bonding:** Attractive dipole interaction between hydrogen covalently bonded to F, O, or N and an electronegative atom.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Bond Order} = \frac{N_b - N_a}{2}$$

Molecular Orbital Bond Order Formula

$$\mu = q \times d \quad (1\ \text{D} = 3.3356 \times 10^{-30}\ \text{C} \cdot \text{m})$$

Electric Dipole Moment of Polar Bond

$$\text{Repulsion: lp-lp} > \text{lp-bp} > \text{bp-bp}$$

VSEPR Electron Pair Repulsion Hierarchy

$$\text{O}_2\ \text{MOT: } \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 (\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^1 = \pi^* 2p_y^1)$$

Electronic Configuration of Paramagnetic O2

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** For homonuclear diatomics with <= 14 electrons (B2, C2, N2), pi 2px = pi 2py orbitals are lower in energy than sigma 2pz!

• **Alert:** NH3 has a much larger dipole moment than NF3 because lone pair and bond pair dipole moments reinforce in NH3 but oppose in NF3.

• **Alert:** Bond order of zero indicates an unstable molecule that cannot exist as a stable chemical species (e.g., He2).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why does NH3 possess a significantly higher dipole moment (1.47 D) than NF3 (0.24 D), even though Fluorine is much more electronegative than Nitrogen? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Dipole Direction in NH3:** Nitrogen is more electronegative than Hydrogen; individual N-H bond dipoles point towards N, reinforcing the lone pair dipole moment in the identical direction. **[1 Mark]**

• **Dipole Opposition in NF3:** Fluorine is more electronegative than Nitrogen; individual N-F bond dipoles point away from N, partially opposing and cancelling the upward lone pair dipole moment. **[1 Mark]**

• **Key Exam Result:** Vector reinforcement in NH3 yields high dipole moment; vector opposition in NF3 lowers it.

2. Define hydrogen bonding. Differentiate between intermolecular and intramolecular hydrogen bonding with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The attractive electrostatic force binding a hydrogen atom covalently attached to a strongly electronegative atom (F, O, N) with another electronegative atom. **[1 Mark]**

• **Types and Examples:** (1) Intermolecular H-bonding occurs between two different molecules (e.g., H2O, HF, or ethanol), raising boiling points. (2) Intramolecular H-bonding occurs within the identical molecule (e.g., o-nitrophenol), lowering boiling points. **[1 Mark]**

• **Key Exam Result:** Intermolecular hydrogen bonding raises viscosity and boiling points in liquid water.

3. Using VSEPR theory, predict the shape, hybridization, and bond angles of: (a) CH4, (b) NH3, (c) H2O, (d) SF6. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **CH4 (Methane):** 4 bond pairs, 0 lone pairs; sp3 hybridized; regular tetrahedral geometry with 109.5 degree bond angles. **[1 Mark]**

• **NH3 and H2O:** NH3 has 3 bp and 1 lp (sp3), trigonal pyramidal with 107 degree angle. H2O has 2 bp and 2 lp (sp3), bent V-shaped with 104.5 degree angle due to strong lp-lp repulsion. **[1 Mark]**

• **SF6 (Sulfur Hexafluoride):** 6 bond pairs, 0 lone pairs; sp3d2 hybridized; regular octahedral geometry with 90 degree bond angles. **[1 Mark]**

• **Key Exam Result:** Repulsion hierarchy lp-lp > lp-bp > bp-bp progressively compresses bond angles.

4. What is hybridization? Describe sp, sp2, and sp3 hybridization with reference to Ethyne (C2H2), Ethene (C2H4), and Methane (CH4). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The phenomenon of intermixing atomic orbitals of slightly different energies to produce entirely new equivalent hybrid orbitals of identical shape and energy. **[1 Mark]**

• **sp and sp2 Hybridization:** In ethyne (C2H2), each carbon undergoes sp hybridization forming 180 degree linear bonds. In ethene (C2H4), each carbon undergoes sp2 hybridization forming 120 degree trigonal planar bonds. **[1 Mark]**

• **sp3 Hybridization:** In methane (CH4), carbon mixes one 2s and three 2p orbitals to form four sp3 hybrid orbitals directed towards tetrahedron corners at 109.5 degrees. **[1 Mark]**

• **Key Exam Result:** Hybridization explains directional valency, molecular geometries, and bond angles.

5. Write the molecular orbital configuration of Oxygen molecule (O2). Calculate its bond order and account for its magnetic behaviour. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Electronic Configuration:** O2 has 16 electrons: sigma 1s2, sigma* 1s2, sigma 2s2, sigma* 2s2, sigma 2pz2, (pi 2px2 = pi 2py2), (pi* 2px1 = pi* 2py1). **[1 Mark]**

• **Bond Order Calculation:** Bonding electrons Nb = 10, Antibonding electrons Na = 6. Bond order = (Nb - Na) divided by 2 = (10 - 6) divided by 2 = 2.0 (Double bond). **[1 Mark]**

• **Paramagnetism Justification:** The presence of two unpaired electrons in degenerate pi* 2px and pi* 2py antibonding molecular orbitals proves that O2 is paramagnetic. **[1 Mark]**

• **Key Exam Result:** MO theory explains the paramagnetism of O2 which valence bond theory failed to justify.

$$\text{Bond Order} = \frac{10-6}{2} = 2, \quad \pi^* 2p_x^1 = \pi^* 2p_y^1 \rightarrow \text{Paramagnetic}$$

Oxygen Molecular Orbital Parameters

6. Write the molecular orbital configuration of Nitrogen molecule (N2). Calculate its bond order and compare the stability of N2 and N2+. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Molecular Orbital Configuration:** N2 has 14 electrons: sigma 1s2, sigma* 1s2, sigma 2s2, sigma* 2s2, (pi 2px2 = pi 2py2), sigma 2pz2. (Note: pi 2p is lower than sigma 2p for <= 14 e-). **[1 Mark]**

• **Bond Order of N2:** Nb = 10, Na = 4. Bond order = (10 - 4) divided by 2 = 3.0 (Triple bond, diamagnetic). **[1 Mark]**

• **Comparison with N2+:** Removing one electron from bonding sigma 2pz gives N2+ with Nb = 9, Na = 4. Bond order of N2+ = (9 - 4) divided by 2 = 2.5. Since higher bond order means greater stability, N2 is more stable than N2+. **[1 Mark]**

• **Key Exam Result:** Bond order of N2 = 3.0; Bond order of N2+ = 2.5; N2 is more stable.

$$\text{Bond Order } (N_2) = \frac{10-4}{2} = 3.0 > \text{Bond Order } (N_2^+) = \frac{9-4}{2} = 2.5$$

Nitrogen Molecular Orbital Bond Order

7. Differentiate between Sigma (σ) and Pi (π) bonds on the basis of: (a) mode of orbital overlap, (b) bond strength, (c) rotation of atoms. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Mode of Overlap:** Sigma bond is formed by end-to-end (axial and head-on) overlap of orbitals along the internuclear axis. Pi bond is formed by lateral (sideways) overlap of parallel p-orbitals perpendicular to the axis. **[1 Mark]**

• **Bond Strength:** Extent of overlap is significantly greater along the internuclear axis, so sigma bond is much stronger than pi bond. **[1 Mark]**

• **Rotation of Atoms:** Free rotation around a single sigma bond is possible without breaking overlap; rotation around a pi bond is restricted and would destroy sideways overlap. **[1 Mark]**

• **Key Exam Result:** Sigma bonds provide molecular framework; pi bonds provide multiple bond reactivity.